

SIMON V3 Research Strengthening: Summary

Six enhancements to novelty, claims, and robustness | April 2026

The following work was conducted after V2 to strengthen the paper's scientific contribution, generalise its results, and sharpen its claims ahead of V3.

1. Architecture Comparison: Proving Analytic Force Direction Is Necessary

V2 claimed SIMON's scalar correction architecture is physically constrained, but never tested what happens when that constraint is removed. A Vector NN baseline was trained with comparable size and strictly more information than SIMON, predicting force direction rather than fixing it analytically. Both models show identical divergence rates during the bounded phase, but the Vector NN causes body ejection at year 35 (final RMS 83.0 AU) while SIMON remains bounded for the full 100 years (1.54 AU). A gate sensitivity analysis confirmed the ejection arises from cumulative 0.04 deg per-step directional errors, not single anomalous predictions. This transforms the paper's contribution from an engineering result into a demonstrated scientific principle: physics-based direction enforcement is necessary, not merely beneficial, for chaotic orbital stability.

2. Multi-Initial-Condition Evaluation: Demonstrating Generalisation

V2 evaluated SIMON on a single initial condition, leaving generalisation claims vulnerable to challenge. SIMON was evaluated on four qualitatively different orbital configurations covering hierarchical, tight-binary, near-equal-mass, and the original V2 case. Bounded orbital dynamics were maintained in all three physically stable configurations (mean divergence rate $\lambda = 0.128 \pm 0.028/\text{yr}$). The one ejecting case was confirmed as a genuine physical instability: the reference integrator ias15 also ejects a body to 58.9 AU for the same configuration, proving SIMON correctly reproduces the true physics rather than introducing a numerical artefact.

3. Energy Conservation Tracking: A Second Independent Stability Metric

V2 had no energy conservation analysis, which is a standard requirement for physics simulation papers. Fractional energy drift was tracked over 100 years for both ias15 and SIMON. The key finding is that SIMON's energy drift (1.39% for the default configuration) is attributable to the leapfrog integrator, not the NN correction. Configurations where the NN was never invoked show comparable drift, proving the NN contributes no additional energy error. This provides a second independent metric corroborating SIMON's physical plausibility and gives the paper a clear, honest answer to the energy conservation question.

4. Adaptive Sub-Stepping: Quantifying the Most Impactful Engineering Decision

V2 described adaptive sub-stepping mechanically but never presented the before/after comparison, leaving the most dramatic result in the project invisible. Running SIMON with and without the mechanism showed: worst-case error drops from 161.7 AU to 1.54 AU (105x reduction), divergence rate drops 38%, and body ejections are eliminated entirely. The cost is 35% more per-step compute time (17.9 to 24.3 us), activating on only 0.2-1.1% of steps. This result directly supports the claim that SIMON produces physically correct results across all tested timesteps, and is now quantified for the first time.

5. Training Seed Robustness: Confirming Reproducibility

V2 reported a single training run, which raises the question of seed dependence. The model was retrained five times with different random seeds and each was evaluated through the full simulation. Training MSE varied by 15.8% across seeds, but all produced identical simulation outcomes: divergence rate $\lambda = 0.1655/\text{yr}$ for every seed, all bodies bounded in every run. The MSE variation is concentrated at sub-softening distances where the safety gate already bypasses NN predictions, making it operationally irrelevant. Results are confirmed reproducible and not the product of a specific initialisation.

6. NN Ablation Study and Claims Refinement: Separating Stability Mechanisms

To precisely characterise what the NN contributes, an ablation was run removing the NN correction while retaining all other mechanisms (adaptive stepping, same timestep). Bodies remained bounded with 8.5% larger final error (1.67 vs 1.54 AU), establishing that adaptive sub-stepping is the primary stability mechanism and the NN provides quantified force accuracy improvement. This replaces V2's over-defensive framing ('NN used only 1.1% of steps') with a precise, evidenced claim. Additional refinements address the speed explanation (separating algorithmic vs hardware contributions), force notation consistency, softening justification, and divergence rate interpretation.

Full details, figures, and exact paper modification text are in the accompanying reports.